P1 - Reliable Batch Pipeline

Portfolio Project 1 | Orders + API + PostgreSQL | RC2

Business scenario

A small marketplace sends order CSV files, customer data through a paginated API, and an operational PostgreSQL table. Build a reproducible batch pipeline that preserves raw data, validates records, reconciles counts/totals, loads trusted tables safely, and can be rerun without logical duplicates.

Required deliverables

- Ingest CSV, paginated API payloads and database extract.
- Preserve immutable raw inputs and batch metadata.
- Reject missing order IDs, invalid dates and negative amounts into quarantine.
- Handle duplicate order rows and duplicate input files deliberately.
- Build PostgreSQL staging/trusted schema and safe load/upsert logic.
- Add logging, config/secrets separation, tests and a reconciliation report.
- Prove rerun idempotency and document a backfill procedure.
- Trigger and repair the supplied partial-API or malformed-record incident.
- Deliver README, architecture, runbook and technical defense.

Deliberate failure pack

- Duplicate order rows
- Missing order_id
- Bad date
- Negative amount
- Repeated input file
- Paginated API
- Malformed API record
- Simulated partial API failure
- Rerun after partial success

Submission rule

Run the local project checker, preserve failure/recovery evidence, and save a final README + architecture + runbook. Live tool/runtime requirements remain conditional where the build environment cannot execute them.